Proteomics analysis of human iPSC-derived cardiac spheroids after laser injury (Sph005)

Differential protein expression and pathway enrichment: Laser vs Control spheroids, 7 days post-laser-injury

2026-08-07

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```
rm(list = ls())
gc()
set.seed(1234)
library(tidyverse)
library(readxl)
source("reusable_functions/source_all.R")
```

A matrix: 2 x 6 of type dbl

	used	(Mb)	gc trigger	(Mb)	max used	(Mb)
Ncells	664414	35.5	1474277	78.8	1121085	59.9
Vcells	1230680	9.4	8388608	64.0	1975134	15.1

```
<e2><9c><93> viz_functions.R loaded: PCA(), draw_volcano()

<e2><9c><93> enrichment_functions.R loaded: ORA_function(), GSEA_function(), run_ORA_loop(), run_GSEA_l

<e2><9c><93> Loaded utils_functions.R
<e2><9c><93> sc_functions.R loaded
  Clustering QC  : compute_clustering_qc(), plot_clustering_qc()
  Annotation     : annotate_celltypes_enrichr()
  DE selection   : compute_data_characteristics(), recommend_de_method()
  DE dispatcher  : run_de_per_celltype()
  DE methods     : run_pseudobulk_limma(), run_limmatrend_cov(),
                   run_limmavoom_cov(), run_edgeR_cov(),
                   run_mast_cov(), run_wilcox_raw()
  LIANA          : run_liana_zebrafish(), run_liana_loop(), plot_liana_dotplot()
  SC enrichment  : run_ORA_loop_sc(), run_GSEA_loop_sc()
<e2><9c><93> All Mercader Lab reusable functions loaded.
  Bulk:        PCA(), draw_volcano(), make_gct(), process_venn_sets()
  Enrichment:  ORA_function(), GSEA_function(), run_ORA_loop(), run_GSEA_loop()
  Orthology:   convert_zf_to_mouse_alliance(), convert_zf_to_human_alliance()
  Single-cell: run_liana_zebrafish(), run_liana_loop(), plot_liana_dotplot(),
               pseudobulk_de(), run_ORA_loop_sc(), run_GSEA_loop_sc()
```

```
# Sph005: human iPSC-derived spheroid laser-injury proteomics (Ctrl vs Laser, 7dpl)
# Read the Spectronaut DE sheet (pre-computed empirical-Bayes stats, contrast = Laser - Ctrl)
combined_data <- read_xlsx("Sph005_Results_Proteomics/20260528_232_RS.xlsx",
                           sheet = "proteinGroups DE test")
```

```r
dim(combined_data)

# Sample metadata
sample_meta <- read_xlsx("OMICS_SampleSet.xlsx", sheet = "Sheet1")
sample_meta
```

1. 4080
2. 131

A tibble: 10 x 6

Short Name <chr>	Long name <chr>	Aggregation <chr>	Group <chr>	Timepoint <chr>	Sample composition <chr>
C1	Control 1	A1	Ctrl	7dpl	24 pooled spheroids
C2	Control 2	B1	Ctrl	7dpl	24 pooled spheroids
C3	Control 3	A2	Ctrl	7dpl	24 pooled spheroids
C4	Control 4	B2	Ctrl	7dpl	24 pooled spheroids
C5	Control 5	A3	Ctrl	7dpl	24 pooled spheroids
L1	Laser 1	A1	Laser	7dpl	24 pooled spheroids
L2	Laser 2	B1	Laser	7dpl	24 pooled spheroids
L3	Laser 3	A2	Laser	7dpl	24 pooled spheroids
L4	Laser 4	B2	Laser	7dpl	24 pooled spheroids
L5	Laser 5	A3	Laser	7dpl	24 pooled spheroids

```r
# Select protein-annotation columns + iIQ (imputed, log-transformed IQ) quant columns
# + the pre-computed empirical-Bayes DE stats for the single Laser - Ctrl contrast
protein_columns <- c("Leading_Protein", "PG.ProteinGroups", "Leading_Gene",
                     "PG.ProteinAccessions", "PG.Genes", "PG.Organisms",
                     "PG.ProteinDescriptions")

iIQ_sample_columns <- colnames(combined_data)[grepl("^iIQ (C|L)[0-9]$", colnames(combined_data))]
iIQ_stat_columns   <- colnames(combined_data)[grepl("^iIQ .*Laser - Ctrl$", colnames(combined_data))]

columns_to_select <- c(protein_columns, iIQ_sample_columns, iIQ_stat_columns)

combined_data_iIQ <- combined_data %>%
  dplyr::select(all_of(columns_to_select))

# sanity check
if (length(iIQ_sample_columns) == 10 && length(iIQ_stat_columns) > 0) {
  print("All iIQ sample + Laser - Ctrl stat columns selected")
} else {
  print("Column selection mismatch - check regex")
}
colnames(combined_data_iIQ)
dim(combined_data_iIQ)
```

```
[1] "All iIQ sample + Laser - Ctrl stat columns selected"
```

1. 'Leading_Protein'
2. 'PG.ProteinGroups'
3. 'Leading_Gene'
4. 'PG.ProteinAccessions'
5. 'PG.Genes'
6. 'PG.Organisms'
7. 'PG.ProteinDescriptions'
8. 'iIQ C1'
9. 'iIQ C2'

10. 'iIQ C3'
11. 'iIQ C4'
12. 'iIQ C5'
13. 'iIQ L1'
14. 'iIQ L2'
15. 'iIQ L3'
16. 'iIQ L4'
17. 'iIQ L5'
18. 'iIQ AND iSN Persistent Signif Laser - Ctrl'
19. 'iIQ log2FC Laser - Ctrl'
20. 'iIQ EB_stat Laser - Ctrl'
21. 'iIQ EB_pVal Laser - Ctrl'
22. 'iIQ EB_adjpVal Laser - Ctrl'
23. 'iIQ Signif Laser - Ctrl'
24. 'iIQ Proportion Signif Laser - Ctrl'
25. 'iIQ Persistent Signif Laser - Ctrl'

1. 4080
2. 25

```
# DEG count for the single Laser - Ctrl contrast (BH-adjusted p < 0.05)
DEG_numbers <- combined_data_iIQ %>%
  summarise(Number_of_DEGs = sum(`iIQ EB_adjpVal Laser - Ctrl` < 0.05, na.rm = TRUE)) %>%
  mutate(Comparison = "Laser - Ctrl", .before = 1)

DEG_numbers
```

A tibble: 1 x 2

Comparison	Number_of_DEGs
<chr>	<int>
Laser - Ctrl	405

```
# Full results table (all proteins, not just significant) with columns renamed to the
# log2FoldChange / padj / gene convention expected by draw_volcano() and GSEA_function()
de_results <- combined_data_iIQ %>%
  dplyr::select(all_of(protein_columns),
                log2FoldChange = `iIQ log2FC Laser - Ctrl`,
                stat           = `iIQ EB_stat Laser - Ctrl`,
                pvalue         = `iIQ EB_pVal Laser - Ctrl`,
                padj           = `iIQ EB_adjpVal Laser - Ctrl`) %>%
  mutate(gene = coalesce(PG.Genes, Leading_Gene))

dim(de_results)
head(de_results)

# Differential proteins at padj < 0.05
differential_proteins_Laser_Ctrl <- de_results %>%
  filter(padj < 0.05)

dim(differential_proteins_Laser_Ctrl)

differential_proteins_Laser_Ctrl %>%
  summarise(upregulated   = sum(log2FoldChange > 0, na.rm = TRUE),
            downregulated = sum(log2FoldChange < 0, na.rm = TRUE))
```

1. 4080
2. 12

A tibble: 6 x 12

Leading_Protein <chr>	PG.ProteinGroups <chr>	Leading_Gene <chr>	PG.ProteinAccessions <chr>	P <c
A0AVT1	A0AVT1	UBA6	A0AVT1	U
A0FGR8-2	A0FGR8;A0FGR8-2;A0FGR8-6	ESYT2	A0FGR8;A0FGR8-2;A0FGR8-6	E
A0JNW5	A0JNW5	BLTP3B	A0JNW5	E
A0PJW6	A0PJW6	TMEM223	A0PJW6	T
A1L0T0	A1L0T0	ILVBL	A1L0T0	I
A1X283	A1X283	SH3PXD2B	A1X283	S

1. 405
2. 12

A tibble: 1 x 2

upregulated <int>	downregulated <int>
85	320

```
# Checkpoint 1: DE table + DEG counts built
save.image(file = "Analysis_Sph005.RData")
```

```
# PCA / sample QC on the iIQ (imputed, log) intensity matrix
iIQ_mat <- combined_data_iIQ %>%
  dplyr::select(all_of(iIQ_sample_columns)) %>%
  as.matrix()
rownames(iIQ_mat) <- combined_data_iIQ$PG.Genes
colnames(iIQ_mat) <- gsub("^iIQ ", "", colnames(iIQ_mat))

# keep only complete-observation proteins for PCA
iIQ_mat_complete <- iIQ_mat[complete.cases(iIQ_mat), ]
dim(iIQ_mat_complete)

# order sample_meta to match matrix column order
sample_meta_ord <- sample_meta[match(colnames(iIQ_mat_complete), sample_meta$`Short Name`), ]

dir.create("QC", showWarnings = FALSE)

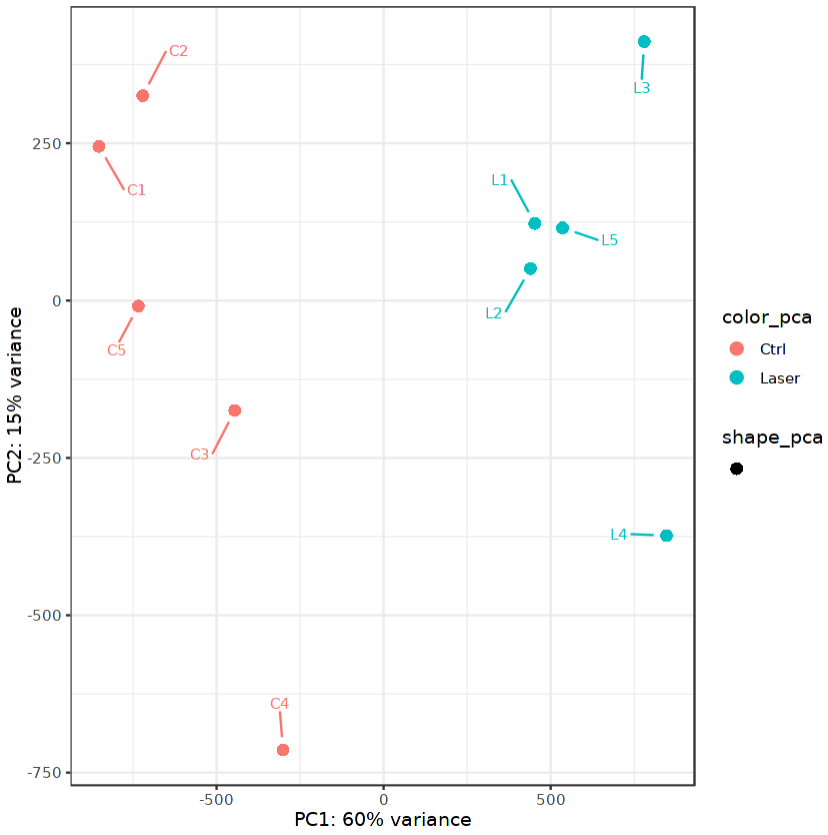
pca_plot <- PCA(iIQ_mat_complete,
    color_pca    = sample_meta_ord$Group,
    label_pca    = sample_meta_ord$`Short Name`,
    save_plot    = "yes",
    name_of_plot = "QC/PCA_Laser_vs_Ctrl")

# PCA() only saves a PDF - also save a 300dpi PNG
ggsave("QC/PCA_Laser_vs_Ctrl.png",
       pca_plot, width = 12, height = 12, dpi = 300)
```

1. 3875
2. 10

```
PCA running...
Percents calculated...
Data frame built...
Plotting...
```

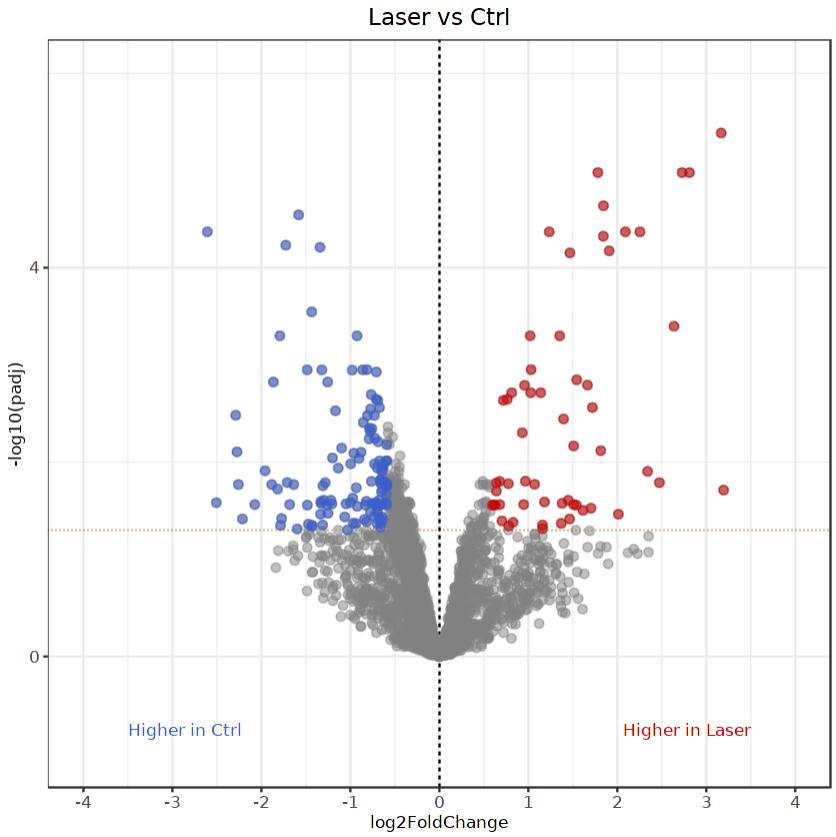
```
Saving plot as: QC/PCA_Laser_vs_Ctrl.pdf
Done
```




```
# Volcano plot: Laser vs Ctrl
dir.create("graphs", showWarnings = FALSE)

volcano_plot <- draw_volcano(de_results, title = "Laser vs Ctrl", lfcthres = 0.585)
volcano_plot

ggsave("graphs/Volcano_Laser_vs_Ctrl_lfcthres_0.585.pdf", volcano_plot, width = 8, height = 7)
ggsave("graphs/Volcano_Laser_vs_Ctrl_lfcthres_0.585.png", volcano_plot, width = 8, height = 7, dpi = 300
```




```
# Checkpoint 2: PCA + volcano figures generated
save.image(file = "Analysis_Sph005.RData")
```

```
# GO / Reactome / GSEA enrichment for Laser vs Ctrl
# Human data - no zebrafish->mouse ortholog conversion needed
CONTRASTS <- "Laser_vs_Ctrl"
LFC_THRESHOLDS <- list(lfc0.585 = 0.585, lfc1 = 1.0, lfc2 = 2.0)   # numeric keys only
ORGANISM <- "human"

# ORA: per-contrast, per-threshold significant (padj < 0.05) gene + log2FoldChange table
filtered_results <- list(
  Laser_vs_Ctrl = lapply(LFC_THRESHOLDS, function(thresh) {
    de_results %>%
      filter(padj < 0.05, abs(log2FoldChange) > thresh) %>%
      dplyr::select(gene, log2FoldChange)
  })
)

# GSEA: ranked results per contrast, restricted to padj < 0.05 significant genes
results_list <- list(Laser_vs_Ctrl = de_results)

run_ORA_loop(filtered_results, contrasts = CONTRASTS, thresholds = LFC_THRESHOLDS, org = ORGANISM)
run_GSEA_loop(results_list, contrasts = CONTRASTS, thresholds = LFC_THRESHOLDS, organism = ORGANISM,
              pval_deg = 0.05)
```

```
Starting GSEA analysis...
Total analyses: 1 <c3><97> 5 <c3><97> 3 = 15
```

```
=== GSEA for contrast: Laser_vs_Ctrl ===

--- Threshold: lfc0.585 ---
  <e2><86><92> BP ... # A tibble: 1 x 3
  total_genes upregulated downregulated
        <int>       <int>         <int>
1         184          57           127
  Ranked gene list: 184 genes
  Detected ID type: SYMBOL

  Mapped: 178 genes to ENTREZID

  <e2><9c><93> GSEA completed: BP - 25 pathways, 15 plotted <e2><86><92> pathway_analysis/GSEA/lfc0.585/
  <e2><86><92> MF ... # A tibble: 1 x 3
  total_genes upregulated downregulated
        <int>       <int>         <int>
1         184          57           127
  Ranked gene list: 184 genes
  Detected ID type: SYMBOL

  Mapped: 178 genes to ENTREZID

  <e2><9c><93> GSEA completed: MF - 23 pathways, 15 plotted <e2><86><92> pathway_analysis/GSEA/lfc0.585/
  <e2><86><92> CC ... # A tibble: 1 x 3
  total_genes upregulated downregulated
        <int>       <int>         <int>
1         184          57           127
  Ranked gene list: 184 genes
  Detected ID type: SYMBOL

  Mapped: 178 genes to ENTREZID

  <e2><9c><93> GSEA completed: CC - 42 pathways, 15 plotted <e2><86><92> pathway_analysis/GSEA/lfc0.585/
  <e2><86><92> reactome ... # A tibble: 1 x 3
  total_genes upregulated downregulated
        <int>       <int>         <int>
1         184          57           127
  Ranked gene list: 184 genes
  Detected ID type: SYMBOL

  <e2><9c><93> GSEA completed: reactome - 2 pathways, 2 plotted <e2><86><92> pathway_analysis/GSEA/lfc0
  <e2><86><92> msigdb ... # A tibble: 1 x 3
  total_genes upregulated downregulated
        <int>       <int>         <int>
1         184          57           127
  Ranked gene list: 184 genes
  Detected ID type: SYMBOL

  <e2><9c><93> GSEA completed: msigdb - 2 pathways, 2 plotted <e2><86><92> pathway_analysis/GSEA/lfc0.58

--- Threshold: lfc1 ---
  <e2><86><92> BP ... # A tibble: 1 x 3
  total_genes upregulated downregulated
        <int>       <int>         <int>
1          86          39            47
  Ranked gene list: 86 genes
```

```
  Detected ID type: SYMBOL

  Mapped: 85 genes to ENTREZID

  <e2><9c><93> GSEA completed: BP - 15 pathways, 15 plotted <e2><86><92> pathway_analysis/GSEA/lfc1/Las
  <e2><86><92> MF ... # A tibble: 1 x 3
  total_genes upregulated downregulated
        <int>       <int>         <int>
1          86          39            47
  Ranked gene list: 86 genes
  Detected ID type: SYMBOL

  Mapped: 85 genes to ENTREZID

  <e2><9c><97> No GSEA enrichment found for MF
  <e2><86><92> CC ... # A tibble: 1 x 3
  total_genes upregulated downregulated
        <int>       <int>         <int>
1          86          39            47
  Ranked gene list: 86 genes
  Detected ID type: SYMBOL

  Mapped: 85 genes to ENTREZID

  <e2><9c><93> GSEA completed: CC - 14 pathways, 14 plotted <e2><86><92> pathway_analysis/GSEA/lfc1/Las
  <e2><86><92> reactome ... # A tibble: 1 x 3
  total_genes upregulated downregulated
        <int>       <int>         <int>
1          86          39            47
  Ranked gene list: 86 genes
  Detected ID type: SYMBOL

  <e2><9c><93> GSEA completed: reactome - 1 pathways, 1 plotted <e2><86><92> pathway_analysis/GSEA/lfc1/
  <e2><86><92> msigdb ... # A tibble: 1 x 3
  total_genes upregulated downregulated
        <int>       <int>         <int>
1          86          39            47
  Ranked gene list: 86 genes
  Detected ID type: SYMBOL

  <e2><9c><97> No GSEA enrichment found for msigdb


--- Threshold: lfc2 ---
  <e2><86><92> BP ... # A tibble: 1 x 3
  total_genes upregulated downregulated
        <int>       <int>         <int>
1          17          10             7
  Ranked gene list: 17 genes
  Detected ID type: SYMBOL

  Mapped: 17 genes to ENTREZID

  <e2><9c><97> No GSEA enrichment found for BP
  <e2><86><92> MF ... # A tibble: 1 x 3
  total_genes upregulated downregulated
        <int>       <int>         <int>
1          17          10             7
  Ranked gene list: 17 genes
  Detected ID type: SYMBOL
```

```
  Mapped: 17 genes to ENTREZID

  <e2><9c><93> GSEA completed: MF - 2 pathways, 2 plotted <e2><86><92> pathway_analysis/GSEA/lfc2/Laser
  <e2><86><92> CC ... # A tibble: 1 x 3
  total_genes upregulated downregulated
        <int>       <int>         <int>
1          17          10             7
  Ranked gene list: 17 genes
  Detected ID type: SYMBOL

  Mapped: 17 genes to ENTREZID

  <e2><9c><97> No GSEA enrichment found for CC
  <e2><86><92> reactome ... # A tibble: 1 x 3
  total_genes upregulated downregulated
        <int>       <int>         <int>
1          17          10             7
  Ranked gene list: 17 genes
  Detected ID type: SYMBOL

  <e2><9c><97> No GSEA enrichment found for reactome
  <e2><86><92> msigdb ... # A tibble: 1 x 3
  total_genes upregulated downregulated
        <int>       <int>         <int>
1          17          10             7
  Ranked gene list: 17 genes
  Detected ID type: SYMBOL

  <e2><9c><97> No GSEA enrichment found for msigdb

<e2><9c><93> Completed all GSEA analyses for Laser_vs_Ctrl

<e2><9c><93><e2><9c><93><e2><9c><93> All GSEA analyses completed <e2><9c><93><e2><9c><93><e2><9c><93>
```

```
# Restore analysis state from checkpoint (fresh kernel session)
source("reusable_functions/source_all.R")
load("Analysis_Sph005.RData")
ls()
```

```
  viz_functions.R loaded: PCA(), draw_volcano()

  enrichment_functions.R loaded: ORA_function(), GSEA_function(), run_ORA_loop(), run_GSEA_loop()

  Loaded utils_functions.R
  sc_functions.R loaded
   Clustering QC  : compute_clustering_qc(), plot_clustering_qc()
   Annotation     : annotate_celltypes_enrichr()
   DE selection   : compute_data_characteristics(), recommend_de_method()
   DE dispatcher  : run_de_per_celltype()
   DE methods     : run_pseudobulk_limma(), run_limmatrend_cov(),
                    run_limmavoom_cov(), run_edgeR_cov(),
                    run_mast_cov(), run_wilcox_raw()
   LIANA          : run_liana_zebrafish(), run_liana_loop(), plot_liana_dotplot()
   SC enrichment  : run_ORA_loop_sc(), run_GSEA_loop_sc()
  All Mercader Lab reusable functions loaded.
   Bulk:        PCA(), draw_volcano(), make_gct(), process_venn_sets()
   Enrichment:  ORA_function(), GSEA_function(), run_ORA_loop(), run_GSEA_loop()
   Orthology:   convert_zf_to_mouse_alliance(), convert_zf_to_human_alliance()
```

```
Single-cell: run_liana_zebrafish(), run_liana_loop(), plot_liana_dotplot(),
             pseudobulk_de(), run_ORA_loop_sc(), run_GSEA_loop_sc()
```

1. 'DEG_numbers'
2. 'GSEA_function'
3. 'ORA_function'
4. 'PCA'
5. 'annotate_celltypes_enrichr'
6. 'base_dir'
7. 'columns_to_select'
8. 'combined_data'
9. 'combined_data_iIQ'
10. 'compute_clustering_qc'
11. 'compute_data_characteristics'
12. 'convert_zf_to_human_alliance'
13. 'convert_zf_to_mouse_alliance'
14. 'de_results'
15. 'differential_proteins_Laser_Ctrl'
16. 'draw_volcano'
17. 'export_table'
18. 'iIQ_mat'
19. 'iIQ_mat_complete'
20. 'iIQ_sample_columns'
21. 'iIQ_stat_columns'
22. 'make_gct'
23. 'pca_plot'
24. 'plot_clustering_qc'
25. 'plot_liana_dotplot'
26. 'process_venn_sets'
27. 'protein_columns'
28. 'pseudobulk_de'
29. 'recommend_de_method'
30. 'run_GSEA_loop'
31. 'run_GSEA_loop_sc'
32. 'run_ORA_loop'
33. 'run_ORA_loop_sc'
34. 'run_de_per_celltype'
35. 'run_edgeR_cov'
36. 'run_liana_loop'
37. 'run_liana_zebrafish'
38. 'run_limmatrend_cov'
39. 'run_limmavoom_cov'
40. 'run_mast_cov'
41. 'run_pseudobulk_limma'
42. 'run_wilcox_raw'
43. 'sample_meta'
44. 'sample_meta_ord'
45. 'top20'
46. 'volcano_plot'
47. 'volcano_top20_plot'

```r
# Excel export: protein identifiers/names + iIQ-only quantification and DE stats
dir.create("tables", showWarnings = FALSE)

export_table <- combined_data_iIQ %>%
```

```
  dplyr::select(-`iIQ AND iSN Persistent Signif Laser - Ctrl`) %>%  # cross-method column, not iIQ-only
  arrange(`iIQ EB_adjpVal Laser - Ctrl`)

writexl::write_xlsx(export_table, "tables/DE_results_iIQ_Laser_vs_Ctrl.xlsx")

dim(export_table)
colnames(export_table)
```

1. 4080
2. 24

1. 'Leading_Protein'
2. 'PG.ProteinGroups'
3. 'Leading_Gene'
4. 'PG.ProteinAccessions'
5. 'PG.Genes'
6. 'PG.Organisms'
7. 'PG.ProteinDescriptions'
8. 'iIQ C1'
9. 'iIQ C2'
10. 'iIQ C3'
11. 'iIQ C4'
12. 'iIQ C5'
13. 'iIQ L1'
14. 'iIQ L2'
15. 'iIQ L3'
16. 'iIQ L4'
17. 'iIQ L5'
18. 'iIQ log2FC Laser - Ctrl'
19. 'iIQ EB_stat Laser - Ctrl'
20. 'iIQ EB_pVal Laser - Ctrl'
21. 'iIQ EB_adjpVal Laser - Ctrl'
22. 'iIQ Signif Laser - Ctrl'
23. 'iIQ Proportion Signif Laser - Ctrl'
24. 'iIQ Persistent Signif Laser - Ctrl'

```
# DEP_summary: counts of differential/upregulated/downregulated proteins (padj < 0.05)
# at increasing |log2FC| effect-size thresholds
DEP_summary <- tibble(LFC_threshold = c(0, 0.585, 1, 2)) %>%
  rowwise() %>%
  mutate(
    Differential  = sum(de_results$padj < 0.05 & abs(de_results$log2FoldChange) > LFC_threshold, na.rm =
    Upregulated   = sum(de_results$padj < 0.05 & de_results$log2FoldChange >  LFC_threshold, na.rm = TRU
    Downregulated = sum(de_results$padj < 0.05 & de_results$log2FoldChange < -LFC_threshold, na.rm = TRU
  ) %>%
  ungroup()

dir.create("tables", showWarnings = FALSE)
write.csv(DEP_summary, "tables/DEP_summary.csv", row.names = FALSE)

DEP_summary
```

A tibble: 4 × 4

LFC_threshold <dbl>	Differential <int>	Upregulated <int>	Downregulated <int>
0.000	405	85	320
0.585	184	57	127
1.000	86	39	47
2.000	17	10	7

```
# Check ranges to confirm draw_volcano()'s internal capping (|lfc|>7, padj<1e-100) is a no-op here
range(de_results$log2FoldChange, na.rm = TRUE)
range(de_results$padj, na.rm = TRUE)
```

1. -2.60781526441795
2. 3.1950533320151

1. 4.11139571250977e-06
2. 0.999226767903137

```
# Volcano plot with top 20 differential proteins labeled
# Must cross BOTH volcano thresholds (padj < 0.05 AND |log2FC| > 0.585, matching draw_volcano's
# own lfcthres = 0.585 coloring), ranked by smallest padj among those
top20 <- de_results %>%
  filter(padj < 0.05, abs(log2FoldChange) > 0.585) %>%
  arrange(padj) %>%
  slice_head(n = 20)

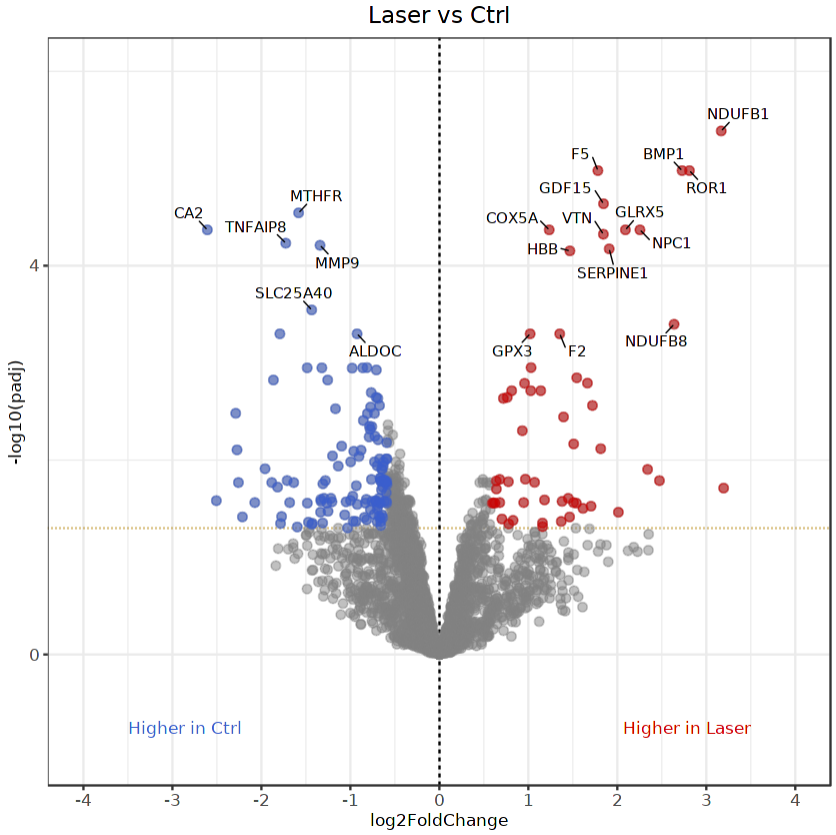
volcano_top20_plot <- volcano_plot +
  ggrepel::geom_text_repel(data = top20,
                           aes(x = log2FoldChange, y = -log10(padj), label = gene),
                           size = 3, max.overlaps = 100, box.padding = 0.4,
                           min.segment.length = 0, segment.size = 0.3)
volcano_top20_plot

ggsave("graphs/Volcano_Laser_vs_Ctrl_top20_labeled_lfcthres_0.585.pdf", volcano_top20_plot, width = 9, h
ggsave("graphs/Volcano_Laser_vs_Ctrl_top20_labeled_lfcthres_0.585.png", volcano_top20_plot, width = 9, h

top20 %>% dplyr::select(gene, log2FoldChange, padj)
```

A tibble: 20 x 3

gene <chr>	log2FoldChange <dbl>	padj <dbl>
NDUFB1	3.1673043	4.111396e-06
F5	1.7817891	1.050613e-05
BMP1	2.7277444	1.050613e-05
ROR1	2.8104685	1.050613e-05
GDF15	1.8446469	2.304476e-05
MTHFR	-1.5815277	2.859750e-05
NPC1	2.2543987	4.276569e-05
CA2	-2.6078153	4.276569e-05
COX5A	1.2345681	4.276569e-05
GLRX5	2.0908309	4.276569e-05
VTN	1.8431685	4.743701e-05
TNFAIP8	-1.7257188	5.865697e-05
MMP9	-1.3409712	6.161943e-05
SERPINE1	1.9085083	6.711795e-05
HBB	1.4672879	7.036361e-05
SLC25A40	-1.4343912	2.849712e-04
NDUFB8	2.6377219	4.004322e-04
F2	1.3531120	5.022464e-04
ALDOC	-0.9248045	5.022464e-04
GPX3	1.0201023	5.022464e-04




```
# Final checkpoint: DE table, figures, enrichment, Excel export, DEP_summary, top-20 volcano all comple
save.image(file = "Analysis_Sph005.RData")
```

```
load("Analysis_Sph005.RData")
```